THEME 2: Environment Data Analytics (EDA)

TASK ID EDA-1: Corporate Carbon Footprint Tracker

Objective:

Collect Scope 1, Scope 2, and Scope 3 emission data from company sustainability disclosures and public filings.

ML/Data Analytics Objective:

- Classify companies based on carbon intensity.
- Predict emission reduction trends.
- Compare actual vs. pledged emission cuts.

Data Extraction Techniques:

- API access (CDP, SEC EDGAR).
- PDF table extraction (sustainability reports) using tabula-py / pdfplumber.

Example Features to Extract: Company name, sector, Scope 1 emissions, Scope 2 emissions, Scope 3 emissions, year, revenue, carbon intensity per revenue.

Deliverables 1:

- Secondary Dataset (CSV) with company-level emissions.
- Python notebooks for API calls + PDF parsing.
- Documentation of extraction methods.

Deliverables 2:

- Python notebooks for classification, trend analysis, and forecasting.
- Documentation of ML methods.
- Automated pipeline to fetch new disclosures and update the dataset.